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Differentiating Breast Tissue Stiffness With Magnetic Resonance Elastography (MRE): A Focus on Fibroglandular and Fatty Tissues

Areej S. Aloufi¹  | Abdullah H. Abujamea² | Faisal S. Fakhouri³  | Salman M. Albeshan¹ | Nora A. Alohal¹ | Reem A. Sabir² | Abeer M. Aldawsari¹ | Reham S. Altokhais¹ | Mehreen A. Malik² | Ashwag R. Alruwaili¹ 

¹Radiological Sciences Department, College of Applied Medical Sciences, King Saud University, Riyadh, Saudi Arabia | ²Department of Radiology and Medical Imaging, College of Medicine, King Saud University, Riyadh, Saudi Arabia | ³Department of Biomedical Technology, College of Applied Medical Sciences, King Saud University, Riyadh, Saudi Arabia

Correspondence: Areej S. Aloufi (aaloufi@ksu.edu.sa)

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ABSTRACT

Breast density is a recognized risk factor for breast cancer and can affect the sensitivity of mammography. Consequently, magnetic resonance imaging (MRI) is recommended as a screening modality for women with increased breast density. However, mammography remains the primary method for assessing a woman's breast density classification. magnetic resonance elastography (MRE) is a new technique to evaluate tissue stiffness characteristics. This study aims to evaluate the ability of MRE to distinguish between fibroglandular and fatty tissues in normal women with different breast densities, potentially aiding in the classification of breast density using MRI. Forty-three women aged 40–79 years with normal screening mammograms were included in this prospective study. MRE was performed using a 1.5-T MRI scanner, and an in-house passive driver was used to obtain an MRE-capable gradient echo (GRE) sequence, which was integrated into a noncontrast-enhanced breast MRI protocol. MRE images were analyzed to measure stiffness values for fibroglandular and fatty tissue based on regions of interest (ROIs) in both breasts. Differences in mean stiffness between tissue types were assessed; a p -value < 0.05 was considered significant. Fibroglandular tissue exhibited significantly higher stiffness than fatty tissue in both breasts (right breast: 1.55 ± 0.31 kPa vs. 0.82 ± 0.13 kPa, $p < 0.001$; left breast: 1.46 ± 0.23 kPa vs. 0.81 ± 0.11 kPa, $p < 0.001$). Comparison between dense and nondense groups on mammograms revealed no significant difference in stiffness for the same tissue types in both breasts. MRE can potentially differentiate between fibroglandular and fatty breast tissues based on shear stiffness, independent of mammographic density. Future research with larger cohorts, including cancer cases, is needed to further establish MRE's role in breast cancer screening.

Abbreviations: ACR, American College of Radiology; BI-RADS, breast imaging reporting and data system; BMI, body mass index; BRCA, breast cancer gene mutation; DENSE, dense tissue and early breast neoplasm screening trial (from citation); DBT, digital breast tomosynthesis; DWI, diffusion-weighted imaging; FOV, field of view; GRE, gradient echo (MRI sequence); HRT, hormone replacement therapy; IDEAL-IQ, iterative decomposition of water and fat with echo asymmetry and least squares estimation quantification; IQR, interquartile range; IRB, Institutional Review Board; kPa, kilopascal; MMDI, multimodel direct inversion; MRE, magnetic resonance elastography; ROI, region of interest; SD, standard deviation; TE, echo time; TR, repetition time.

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1 | Introduction

Breast density, which refers to the relative distribution of fibroglandular and fatty tissue, is a key factor in breast cancer screening and risk assessment [1, 2]. On mammograms, fibroglandular tissue appears white, while fat appears darker. The American College of Radiology (ACR) has developed the breast imaging reporting and data system (BI-RADS), a standardized classification system for categorizing breast density based on the extent of fibroglandular tissue visible on mammograms. Radiologists generally use this system to subjectively classify breast density into four categories (ACR A-D) based on the ratio of dense to fatty tissue [3].

High breast density can significantly reduce the sensitivity of mammography by masking tumors, leading to a 40% reduction in cancer detection rates in women with dense breasts [4]. Moreover, mammographic density has been recognized as an independent risk factor for breast cancer. McCormack et al. conducted a meta-analysis showing that women with high mammographic density ($\geq 75\%$) have a 4 to 6 times higher risk of developing breast cancer compared to women with low density ($< 5\%$) [2]. Several studies confirm the relationship between increased breast density and cancer risk [5–7], though a recent review reported lower relative risk, around twofold [8].

Due to the limitations of mammography in women with dense breasts, supplemental screening methods such as ultrasound and breast magnetic resonance imaging (MRI) are often recommended [9, 10]. MRI particularly has been used as a valuable tool for screening women with a higher risk of breast cancer due to breast cancer gene mutation (BRCA) and strong family or personal history, and recently for women with average risk and high mammographic density [11]. Additionally, MRI has been used in breast imaging for assessing changes in breast tissue in women undergoing hormone replacement therapy (HRT) or treatments like tamoxifen, which are known to influence breast density [12].

A recent advancement in breast imaging is magnetic resonance elastography (MRE). MRE is a phase contrast technology that assesses tissue shear stiffness by measuring tissue displacement in response to external mechanical vibrations [13]; the technology has shown promising results in differentiating between malignant and nonmalignant tissue in the breast [14]. MRE estimates shear stiffness (i.e., the overall resistance of tissue to deformation) and elasticity (i.e., the tissue's ability to return to its original shape after deformation) in which both are estimated in kilopascals (kPa) [15]. It was evident that shear stiffness and elasticity were significantly higher in high-risk patients, particularly those with dense breasts, and were strong predictors of breast cancer risk [16]. Significant differences in breast elasticity had also been observed between dense and nondense groups in normal-risk women, although differences in shear stiffness had not been statistically significant [16]. Breast shear stiffness measurements had been found to be reproducible, with a strong correlation between repeated scans [17].

Women with denser breasts, classified using BI-RADS categories C and D, exhibited significantly higher overall stiffness

values than women with less dense breasts, BI-RADS categories A and B. Specifically, the mean shear stiffness ($\pm$ standard deviation, SD) for dense breasts was 0.96 ± 0.13 kPa (center slice) and 0.92 ± 0.13 kPa (all slices), while for nondense breasts, it was 0.85 ± 0.11 kPa (center slice) and 0.83 ± 0.11 kPa (all slices), with statistically significant differences ($p \leq 0.05$) [17]. Findings from previous research suggest that MRE could be used to assess breast shear stiffness and may be useful for differentiating breast tissue types based on density. Nonetheless, to the authors' knowledge, no recent studies have specifically evaluated MRE's ability to differentiate between normal breast tissue types. Establishing baseline stiffness values for normal fibroglandular and fatty tissues is an essential step toward accurately detecting pathological changes in future studies. Therefore, this study aimed to address that gap by assessing whether MRE can provide distinct stiffness values for fibroglandular and fat tissue in a healthy cohort.

2 | Methods

2.1 | Study Population

The Institutional Review Board at King Saud University Medical City approved this prospective single-center study (IRB No: 20-5430). All subjects consented to participate. The study recruited women who underwent routine screening at King Khaled University Hospital (KKUH) and invited them to attend MRI and MRE scans at the same hospital between January 2022 and December 2022. All participants had undergone screening mammography within the previous 6 months or less before the MRI/MRE scans. The inclusion criteria were as follows: (a) women who had normal full mammography and digital breast tomosynthesis DBT exams (2-year follow-ups of a prior normal mammography and/or ultrasound screening exam); (b) between 40 and 79 years old; (c) not pregnant, nor breastfeeding and with no breast implant; (d) not claustrophobic or contraindicated for MRI; and (e) in their full mental capabilities to understand and sign the consent form. Women included in this study were selected randomly from all breast density categories (ACR BI-RADS breast density categories fifth edition) [18]. The exclusion criteria were women who did not have mammography results, women with suspicious mammography results (BI-RADS 3, 4, or 5) [3] and those with a previous biopsy, treatment or surgical intervention, and/or history of breast cancer, in addition to incomplete examinations and technical failures. Patient consent and clinical characteristics and demographics were collected at the time of the scan, such as age, body mass index (BMI), menopause status, history of breastfeeding, number of children, and family history of breast cancer.

2.2 | Image Acquisition

All eligible women were scanned with a 1.5-T MRI scanner (GE Healthcare, Waukesha, Wisconsin, USA) equipped with a pneumatic elastography system (Resoundant Inc., Rochester, MN, USA) and a dedicated 8-channel phased-array breast coil. Scans were conducted between January 2022 and December 2022. Women were placed in a prone foot-first position, and

a conventional breast coil was used. An in-house-built soft passive MRE driver was positioned between the participant's sternum and the breast coil, following the methodology described in two recent studies [16, 17]. The design of this driver was based on established principles from these studies, ensuring mechanical stability and consistent wave propagation across all scans. Its sternum-to-coil positioning provided uniform vibration transmission, as confirmed by homogeneous wave patterns observed in over 95% of participants. An MRE-capable gradient echo (GRE) sequence was integrated into a noncontrast-enhanced breast MRI protocol (scan time = 20 min) for research purposes to avoid switching RF coils or repositioning.

2.3 | Breast MRE

In this study, a GRE-based MRE-enabled sequence was used to acquire estimated numbers of the shear stiffness of the breast. Scan parameters were TR: 60 ms; TE: 25.18 ms; 6 slices with a thickness of 8- and 10-mm spacing between slices, FOV: 490×400 mm²; image resolution: 256×256; flip angle: 30°; 4 MRE phase offsets; mechanical vibration frequency of 60 Hz; motion encoding gradients frequency of X Hz applied on Z-axis (i.e., through slice). The breast MRI protocol included the following sequences: Axial diffusion-weighted imaging (DWI) with *b*-values of 20, 200, 500, and 800, in addition to the iterative decomposition of water and fat with echo asymmetry and least squares estimation quantification sequence (IDEAL-IQ) with a slice thickness of 5 mm and spacing between slices of 5.5 mm.

2.4 | Image Analysis

In general, shear stiffness can be calculated by using simplified Helmholtz's Equation (1):

$$\mu = \rho V_s^2 \quad (1)$$

where μ is the shear stiffness, ρ is the breast density which is equal to 1000 kg/m³, and V_s is the wave speed. The wave speed can be obtained from the mechanical wave frequency (*f*) multiplied by the wavelength of the propagating wave (λ).

Breast shear stiffness maps were calculated for each slice individually by using a multimodel direct inversion algorithm (MMDI). Wave images and elastograms (stiffness maps) were generated automatically for each image slice on the console and provided to postprocessing software (READY View version 13.0; GE Healthcare). Fatty and fibroglandular areas in the right and left breasts for each woman were outlined by an experienced breast radiologist (10 years of experience) using the magnitude map images only. Therefore, regions of interest (ROIs) were drawn first on magnitude images that show the fatty and fibroglandular structures of the breasts only, and then ROIs were copied to the same area on other sequences, including the MRE image. ROIs were constrained to homogeneous tissue areas, visually confirmed on magnitude maps, with a minimum size requirement of 70 mm². The average ROI sizes were 118 and 192 mm² for the right and left fibroglandular areas, respectively, and 100.3 and 98.5 mm² for the right and left fatty areas, respectively. Heterogeneous

regions, such as fat-fibroglandular interfaces, were explicitly excluded. The central slice was used to acquire the four ROIs of each woman; all chosen ROIs of fatty and fibroglandular areas were inside the 95% confidence checker map, as shown by the MRE software; these checker maps identify the area of high-quality data that should be considered in the stiffness assessment. Afterwards, the average stiffness value for each ROI was recorded in kPa, in addition to the standard deviation and ROI size.

2.5 | Statistical Analysis

Sample characteristics and MRE values were calculated using descriptive statistics. The Independent *t*-test was used to determine the statistical difference between fat and fibroglandular tissue stiffness, assessed by MRE for the right and left breast. A post hoc power analysis (GPower 3.1) using the observed effect size (Cohen's *d* = 2.8 for fibroglandular versus fat tissue stiffness) confirmed a statistical power of 99% at an alpha level of 0.05, validating that our sample size was sufficient to achieve the primary study objective. Moreover, the sample was divided into two groups: the nondense group, which includes ACR BI-RADS 5th edition categories A (the breasts are almost entirely fatty) and B (there are scattered areas of fibroglandular density) and the dense group, including categories C (the breasts are heterogeneously dense, which may obscure small masses) and D (the breasts are extremely dense, which lowers the sensitivity of mammography) [3]. To assess if the type of mammographic group (dense and nondense) would affect the ability of MRE to differentiate between tissue types, an independent *t*-test was used for normally distributed values and Mann-Whitney for nonnormally distributed values. Statistical analysis was performed using IBM SPSS Statistics version 27, and *p*-value < 0.05 was considered statistically significant. To determine the optimal stiffness threshold for differentiating fat from fibroglandular tissue based on the study data, receiver operating characteristic (ROC) analysis and the Z-score intersection method were applied (Data S1).

3 | Results

A total of 43 women fulfilled the inclusion criteria and were included in the study (Figure 1). Excluded cases based on technical difficulties were due to no anatomical parts being found within the 95% confidence checker, as seen on the elastogram map.

The sample age ranges between 40 to 67 years (mean ± SD = 54 ± 7). Approximately 53.5% (*n* = 23) were dense breast patients while 46.5% (*n* = 20) were nondense patients included in the study. The majority of the women (90.7%) are postmenopausal (Table 1). Furthermore, only 16.3% of women have a family history of breast cancer (first- and/or second-degree, Table 1).

The results showed that there was a significant difference in shear stiffness between fibroglandular and fat tissue of the right breast (1.55 ± 0.31 kPa vs. 0.82 ± 0.13 kPa; *p* < 0.001). Also, there was a significant difference in shear stiffness between

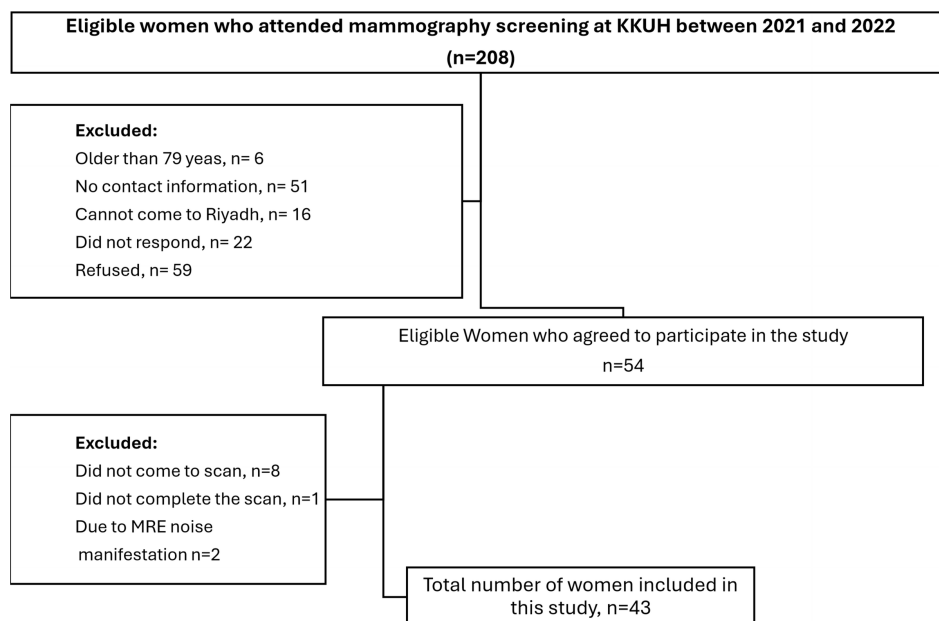


FIGURE 1 | Flowchart of study participants.

TABLE 1 | Clinical and demographic characteristics of study participants ($n = 43$).

Characteristic	N (%)
Age (years)	
Mean (SD)	54 (7)
median (IQR)	55 (10)
BMI (Kg/m²)	
Mean (SD)	31.8 (6.8)
median (IQR)	55 (10)
Mammographic density*	
Nondense groups (A and B)	23 (53.5)
Dense groups (C and D)	20 (46.5)
Menopausal	
Yes	28 (65.1)
No	12 (27.9)
Not known	3 (7)
Family history of breast cancer	
Yes	7 (16.3)
No	36 (83.7)

*American College of Radiology ACR BI-RADS breast density groups 5th edition.

fibroglandular and fat tissue of the left breast (1.46 ± 0.23 kPa vs. 0.81 ± 0.11 kPa; $p < 0.001$) (Table 2).

The total sample result should show a significant difference between fatty and fibroglandular across mean $\pm$ SD (0.82 ± 0.13 kPa vs. 1.51 ± 0.28 kPa; $p < 0.001$), Figure 2 provides a demonstration

of the difference in MRE values of fibroglandular and fat tissue for both breasts.

The calculated stiffness threshold values for differentiating between fat and fibroglandular breast tissue, based on ROC analysis and the Z-score intersection method, were 1.141 and 1.051 kPa, respectively (Data S1).

The results also showed that there is no significant difference between fibroglandular tissue in different density groups for both breast sides; similar findings were reported for fatty tissue (Table 3).

4 | Discussion

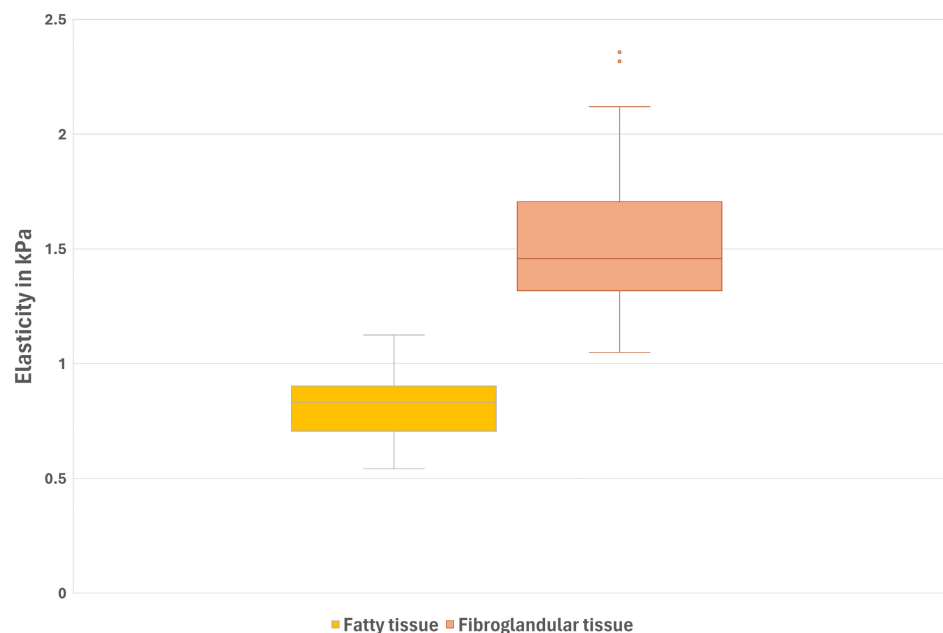
This study demonstrates the potential of MRE as an emerging breast imaging modality to distinguish between fibroglandular and fatty breast tissues, adding to the growing evidence of the utility of MRE in breast imaging. This study aimed to take a step further and investigate the ability of MRE to differentiate fibroglandular tissue from fatty tissue in the breast. The results showed that fibroglandular tissue is significantly stiffer than fatty tissue for both left and right breasts. This study reported shear stiffness values that are comparable to Hawley et al.'s study, where a similar range of stiffness values for dense vs. nondense breast tissue was provided (Figures 3 and 4). In 2017, a study by Hawley et al. reported using breast MRE scans on 22 healthy volunteers with a 3-T MRI. The results of the mean stiffness values of dense breasts, characterized by a higher content of fibroglandular tissue, displayed greater mean shear stiffness values $\pm$ SD (0.96 ± 0.13 kPa for center slices and 0.92 ± 0.13 kPa for all slices) compared to those with nondense breasts (0.85 ± 0.11 kPa for center slices and 0.83 ± 0.11 kPa for all slices) $p \leq 0.05$ [17]. More recently, Patel et al. conducted a study that showed no significant difference between mammographic nondense and dense groups in overall mean tissue stiffness

TABLE 2 | Values of fibroglandular and fat tissues of both breasts based on MRE, ($n = 43$).

Characteristic	Right		<i>p</i> -Value*	Left		<i>p</i> -Value*
	Fibroglandular	Fat		Fibroglandular	Fat	
MRE (kPa)						
Mean $\pm$ SD	1.55 $\pm$ 0.31	0.82 $\pm$ 0.13	<0.001	1.46 $\pm$ 0.23	0.81 $\pm$ 0.11	<0.001
Median $\pm$ IQR	1.47 $\pm$ 0.45	0.82 $\pm$ 0.21		1.45 $\pm$ 0.32	0.83 $\pm$ 0.20	
Maximum	2.35	1.11		2.08	1.04	
Minimum	1.05	0.54		1.12	0.60	

Abbreviations: IQR, interquartile range; kPa, kilopascal; MRE, magnetic resonance elastography; SD, standard deviation.

*Independent *t*-test for normal distribution and Mann–Whitney *U* test for nonnormal distribution.

**FIGURE 2** | Boxplots showing a comparison of MRE stiffness between fibroglandular and fatty tissue for the total sample ($n = 43$).

($\pm$ SD) for average (1.07 ± 0.18 kPa vs. 1.16 ± 0.21 kPa) and high-risk women (1.69 ± 1.19 kPa vs. 2.14 ± 1.12 kPa), $p > 0.05$ [16]. Alternatively, our results suggest that MRE is able to differentiate fibroglandular tissue from fatty tissue ROIs, and this ability is consistent and not affected by side or breast density category. Although hormonal fluctuations may influence breast stiffness, 90.7% of our cohort was postmenopausal (Table 1), minimizing cyclic variations. Future studies should stratify premenopausal participants by menstrual phase. Unlike MRI-based fibroglandular volumetry, which relies on anatomical signal intensity, MRE quantifies functional stiffness. Integrating MRE with MRI could automate density classification and reduce subjectivity in BI-RADS density assessment. Prior research has explored MRE's role in differentiating malignant from benign lesions, highlighting the method's potential to enhance breast cancer screening by providing different stiffness values for different types of tissue in the breast.

In the present study, a direct comparison between MRE-derived stiffness and conventional MRI-based fibroglandular tissue assessment was not performed. This was primarily due to the methodological limitations of our 2D MRE maps,

which restricted analysis to specific ROIs and excluded substantial portions of the breast lying outside the validated regions defined by the 95% confidence checker maps. As such, a whole-breast density analysis comparable to conventional volumetric methods was not feasible. Future work will address this limitation by employing 3D MRE acquisitions and integrating them with additional MRI sequences to enable comprehensive, whole-breast comparisons. These advancements could substantially enhance the value of MRI screening for breast cancer, particularly among high-risk women and those with dense breast tissue [15, 19]. Previous research that investigated breast tissue stiffness in ultrasound using shear wave elastography, particularly, found that malignant lesions generally exhibit much higher stiffness compared to benign and normal tissues, which led to the method being increasingly incorporated into breast ultrasound protocols to enhance diagnostic accuracy [20]. Similarly, in MRE, many studies showed that quantitative stiffness measurements correlate well with histopathological outcomes; the average stiffness of breast carcinoma values was found much higher than the average stiffness of surrounding breast tissues thus aiding in the differentiation between benign and malignant lesions

TABLE 3 | MRE stiffness results (kPa) for fibroglandular and fat tissues of both breasts among the dense and nondense groups.

	Right fibroglandular tissue			Right fat tissue			Left fibroglandular tissue			Left fat tissue		
	Nondense	Dense	p-Value*	Nondense	Dense	p-Value*	Nondense	Dense	p-Value*	Nondense	Dense	p-Value*
Mean $\pm$ SD	1.57 $\pm$ 0.38	1.54 $\pm$ 0.23	0.760	0.81 $\pm$ 0.13	0.85 $\pm$ 0.14	0.337	1.42 $\pm$ 0.23	1.52 $\pm$ 0.24	0.177	0.83 $\pm$ 0.12	0.80 $\pm$ 0.12	0.428
Median (IQR)	1.40 (0.54)	1.59 (0.40)		0.81 (0.20)	0.85 (0.20)		1.44 (0.23)	1.52 (0.32)		0.84 (0.18)	0.80 (0.20)	
Max	2.36	1.95		1.08	1.11		2.08	1.98		1.12	0.98	
Min	1.05	1.14		0.54	0.59		1.10	1.05		0.65	0.61	

Abbreviations: IQR, interquartile range; kPa, kilopascal; MRE, magnetic resonance elastography; SD, standard deviation.

*Independent t-test for normal distribution and Mann-Whitney U test for non-normal distribution.

[15, 19]. One study evaluated six healthy volunteers and found that the mean shear stiffness of breast carcinoma was 418% higher than the average stiffness of surrounding breast tissues [21]. Another study showed that among 35 patients with 20 lesions, the median (range) elasticity in kPa of benign lesions was seven [2–8] compared to 15.9 (8–28) for malignant lesions [22]. Other studies found that MRE increases the diagnostic accuracy of MRI [23, 24].

MRE has also been previously investigated in assessing global breast tissue stiffness. Hawley et al. and Patel et al. assessed MRE differences in overall stiffness values of breast tissue across mammographic breast density categories, more specifically between ACR BI-RADS nondense groups (A and B) and dense groups (C and D) [16, 17]. Both studies showed that MRE provides different elasticity results for different mammographic density groups, although it was more significant in Hawley et al. than in Patel et al.

Other studies have shown that malignant breast lesions exhibit significantly higher stiffness values. For instance, a study from 2002 reported a median stiffness of up to 33 kPa (range: 18–94 kPa) for malignant tissue, compared to a mean $\pm$ SD of 7.5 $\pm$ 3.6 kPa for fibroglandular tissue in healthy volunteers and much lower values for fat tissue at 3.3 $\pm$ 1.9 kPa [21]. The difference in stiffness between normal and malignant breast tissue, where breast cancer typically presents with higher stiffness values, has been consistently observed in subsequent studies. However, shear stiffness values for different breast tissue types are generally lower in recent studies utilizing advanced MRE technology, including the findings of this study [15]. The distinction in stiffness among fibroglandular, fat, and malignant tissue suggests that breast MRE could be employed to establish specific thresholds on elastogram maps to differentiate tissue types. Additionally, MRE could facilitate the automated calculation of breast density as a percentage by area or volume, akin to automated mammography software such as Volpara [25]. Beyond density assessment, MRE holds the potential to highlight suspicious areas indicative of malignant changes, providing structural insights that enhance the accuracy of early breast cancer detection and diagnosis.

This study has several limitations. The results could be stronger if a larger sample size including both cancer and noncancer cases were incorporated, providing a more direct comparison of all tissue types within the same sample. Additionally, our cohort was predominantly postmenopausal (90.7%), as we aimed to investigate women undergoing routine mammography, a population largely composed of postmenopausal women. This limits insights into premenopausal populations; therefore, future studies will target younger women to better assess the hormonal influences on breast stiffness.

The 2D MRE maps generated in this study setting limited the ability to select larger, more comprehensive ROIs due to the nonlinear propagation of waves in the breast. Nevertheless, all included ROIs were chosen from areas within a 95% confidence checker and demonstrated appropriate wave propagations, as confirmed on elastograms and wave maps. Although this approach aimed to minimize bias, it may still introduce selection bias in ROI placement, potentially affecting the study's findings.

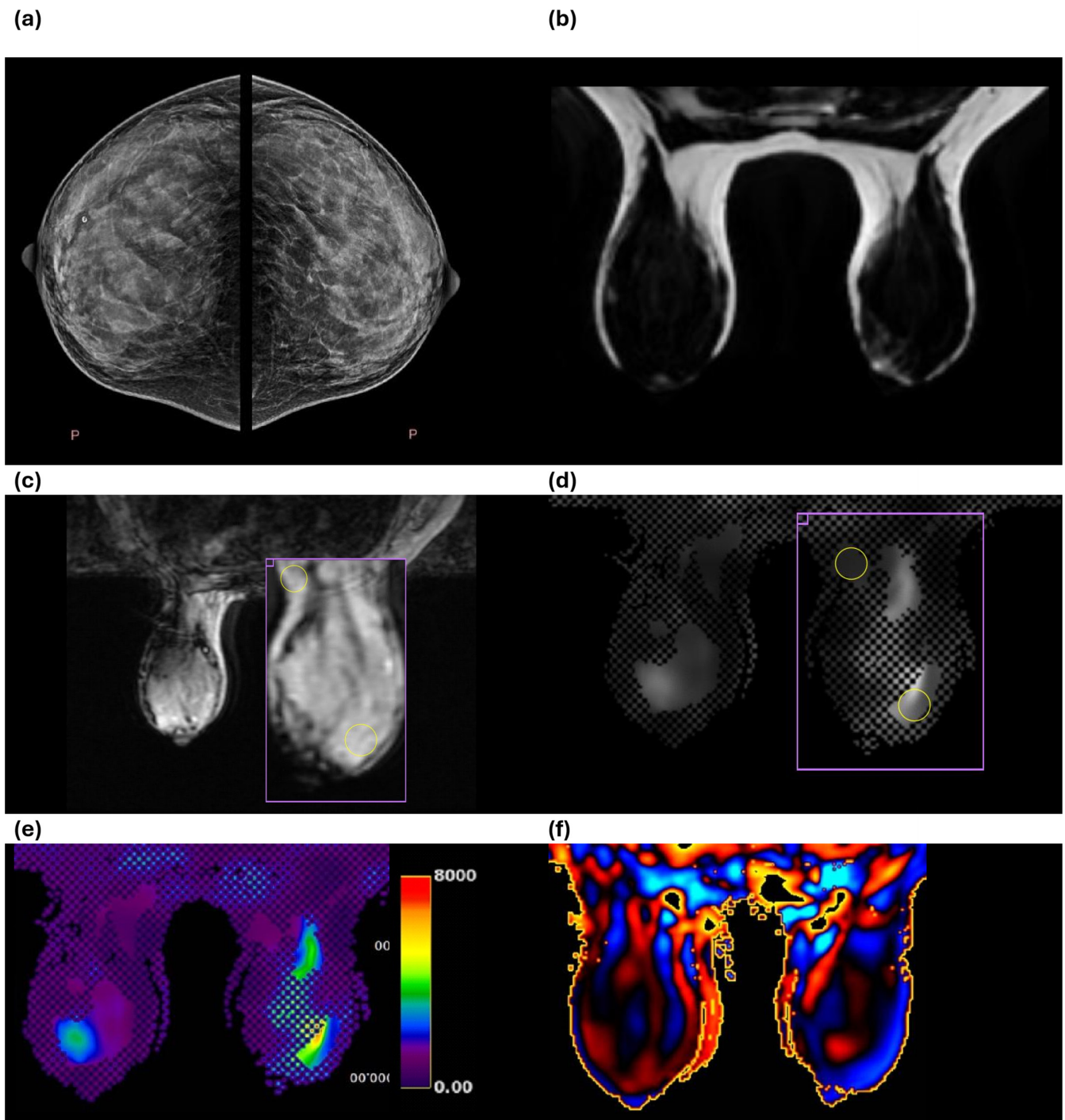


FIGURE 3 | A 43-year-old premenopausal woman with no family history of breast cancer and a BMI of 27.59 kg/m². (a) Mammographic image showing breast density category D (extremely dense breasts). (b) Axial IDEAL-IQ MRI image showing fibroglandular and fatty areas. (c) MR magnitude map demonstrates anatomical anatomy to delineate fibroglandular and fatty areas in the central slice. (d) and (e) are the corresponding elastogram (0–8 kPa) with a 95% confidence checker map, ROIs are seen in images (c and d) where shear stiffness values for a fibroglandular and fatty areas in the right (stiffness values were 1.59 and 0.84 kPa, respectively) and left (stiffness values were 1.82 and 0.65 kPa, respectively) breasts. (f) Snapshot of the wave-processed images showing linear propagating wave areas through the central slice.

Providing access to the entire breast area would allow for a more comprehensive analysis, reducing selection bias and improving the generalizability of the results. All ROIs in this study were selected by the radiologist before reviewing the MRE stiffness map, minimizing potential bias. However, ROIs were placed by a single radiologist, and future work will incorporate interrater

reliability assessments (e.g., intraclass correlation) across multiple observers.

Moreover, a correlation between MRE findings and conventional MRI-based fibroglandular tissue (FGT) assessment was not performed. While such correlation could have provided additional

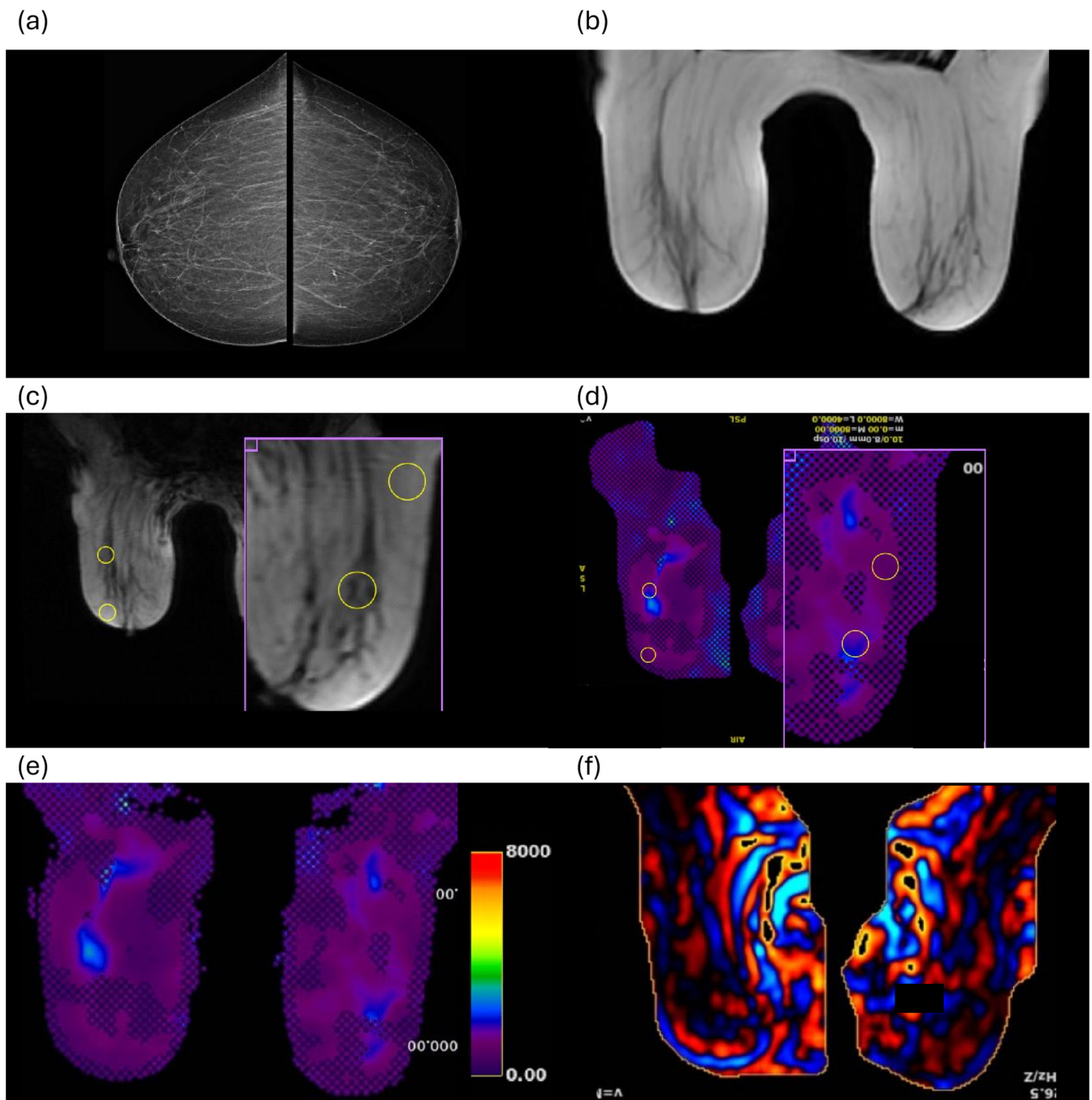


FIGURE 4 | A 65 women postmenopausal women with no family history of breast cancer and a BMI of 36.92 kg/m². (a) mammographic image showing breast density category A entirely fatty breasts. (b) FAT: Ax IDEAL-IQ image showing fibroglandular and fatty areas. (c) MR magnitude map demonstrates anatomical anatomy to delineate fibroglandular and fatty areas in the central slice. (d) and (e) are the corresponding elastogram (0–8 kPa) with a 95% confidence checker, ROIs are seen in images c and d where stiffness values for fibroglandular and fatty areas in the right (stiffness values were 1.48 and 0.81 kPa, respectively) and left (stiffness values were 1.48 and 0.67 kPa, respectively) breasts. (f) snapshot of the wave showing linear propagating waves through the central slice.

validation against established breast tissue categorization methods, it was not feasible in our study due to methodological constraints. Conventional breast density evaluation in clinical and screening practice is based on an overall breast density assessment, whereas our approach focused on specific ROIs within the breast tissue. Furthermore, the use of 2D elasticity maps limited the analysis, as substantial portions of tissue lay outside

the validated areas defined by the 95% confidence checker maps, making a comprehensive overall assessment impractical.

The use of a 1.5-T setup and an in-house driver may limit generalizability; however, the observed stiffness values were consistent with those reported in studies conducted at 3 T [17], supporting the robustness of our measurements [16]. We recommend the

use of standardized commercial passive drivers for future multicenter validation.

The identified stiffness thresholds of 1.141 and 1.051 kPa for distinguishing fat from fibroglandular tissue suggest that MRE holds promise for tissue characterization in the breast; however, these findings represent preliminary results, and larger studies are needed to confirm and validate these thresholds. Future work should also explore integrating MRE into multiparametric MRI protocols (e.g., combining DWI, T2 mapping, and MRE) to enhance cancer risk stratification. Developing automated approaches based on stiffness maps may support rapid screening and diagnosis and facilitate personalized diagnostic strategies. The ability of MRE to distinguish tissue types and detect subtle changes in stiffness may open doors for the integration into breast cancer screening protocols, particularly for individuals with dense breasts or at high risk. By leveraging these objective metrics, MRE has the potential to provide a standardized and reproducible method for assessing breast density, eliminating the subjectivity associated with mammographic ACR BI-RADS classifications.

5 | Conclusion

This study highlights the potential of MRE as a noninvasive imaging modality for differentiating fibroglandular and fatty breast tissues, providing consistent and significant stiffness differences in both breasts. There were no notable differences in fibroglandular tissue stiffness values and similarly for fatty tissues according to mammographic density categories, dense vs. nondense groups. This suggests that MRE's capacity to differentiate tissue types is not influenced by mammographic density. In clinical practice and breast cancer screening settings, MRE could be utilized to enhance tissue characterization, potentially aiding in the assessment of breast cancer risk and improving targeted screening strategies.

Author Contributions

Areej S. Aloufi was responsible for conceptualization, data collection, data cleaning, statistical analysis, and drafting the first version of the manuscript. Abdullah H. Abujamea and Faisal S. Fakhouri contributed to the study and protocol design, image acquisition, methodology development, and critically reviewed and edited the manuscript. Salman M. Albeshan participated in patient recruitment and manuscript revision. Nora A. Alohalay, Abeer M. Aldawsari, and Reham S. Altokhais contributed to patient recruitment and data collection. Reem A. Sabir was involved in image acquisition and data collection. Mehreen A. Malik contributed to data collection, image analysis, and supervision. Ashwag R. Alruwaili participated in image acquisition, data collection, manuscript revision, and supervision. All authors read and approved the final version of the manuscript.

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Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Receiver operating characteristic (ROC) curve for differentiating fat and fibroglandular tissue using magnetic resonance elastography (MRE) stiffness values. The curve illustrates the trade-off between sensitivity (true positive rate) and 1-specificity (false positive rate). The blue shaded region represents the area under the curve ($AUC = 0.997$), indicating excellent discriminative performance. The red dot marks the optimal threshold (≈ 1.07 kPa), determined using Youden's Index, where sensitivity and specificity are maximized. The dashed diagonal line represents chance-level performance ($AUC = 0.5$).